MIL-S-36908A
16 September 1976
SUPERSEDING
MIL-S-36908 (DSA-DM)
31 October 1972

MILITARY SPECIFICATION

SHAKING MACHINE, LABORATORY, Single Cup, 115 volt, 50-60 Hz, AC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers a variable speed laboratory mixer for use in mixing small quantities of materials in hand-held tubes or small flasks.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-M-7298	- Manual, Technical: Commercial Equipment.

Benefical comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640

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STANDARDS

FEDERAL

FED-STD-601 - Rubber: Sampling and Testing

MILITARY

MIL-STD-105 - Sampling, Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storages

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73-11 Dimensions of Plugs and Receptacles-General Purpose, 125 Volts, 15 Amperes, 2 Pole, 3-Wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

UNDERWRITERS' LABORATORIES, INC.

UL 73 Motor-Operated Appliances.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, L.I., NY 11746; 207 East Ohio Street, Chicago, IL 60611; and 1655 Scott Boulevard, Santa Clara, CA 95050.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable

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requirements of UL 73 for motor-operated appliances. Proof shall consist of one of the following:

a. UL label or UL listing of the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the contracting officer, upon the latter's request.

b. Laboratory report from a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government which reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified.

3.1.2 Compliance with above paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.2 Material.

3.2.1 Base. The base shall be either die-cast aluminum with a minimum wall thickness of 1/16 inch, or cast iron with a minimum wall thickness of 3/16 inch.

3.2.2 Sponge strips. The sponge strips shall be rubber or neoprene.

* 3.2.3 Cup receptacle. The cup receptacle shall be neoprene, isoprene or polyurethane. When tested in accordance with 4.4.3, the material shall have a durometer hardness of 50-55 for neoprene and isoprene, or 70-75 for polyurethane.

3.2.4 Electrical components. The electrical components (e.g., motor, switches, rheostat, line cord and wiring) shall meet the requirements of 3.1 and 3.3. The current rating for any component shall be not less than the current rating for the laboratory mixer.

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3.3 Components.

3.3.1 Cup receptacle. A cup receptacle shall be provided and shall have a minimum opening diameter of 1 inch and a minimum depth of 3/4 inch. The bottom of the cavity shall have tapered sides. The cavity of the cup receptacle shall have a minimum wall thickness of 3/16 inch.

* 3.3.2 Sponge strips. Two sponge strips shall be provided and shall be at least 3/4 inch wide x 1/4 inch thick.

3.3.3 Electrical components.

3.3.3.1 Motor. An oilless-bearing induction type motor shall be provided and shall be rated for operation 115 ± 5 volts, 50/60 Hz, AC.

3.3.3.2 Rheostat. A wire-wound rheostat shall be provided with a control or pointer knob. The rheostat shall be provided with a dial plate marked to indicate fixed settings for speed control. The settings shall not be required to be calibrated.

3.3.3.3 On-off switch. An on-off switch shall be provided and shall be either a toggle or push-button type.

3.3.3.4 Pressure-actuated switch. A pressure-actuated switch shall be provided and shall be either a microswitch type, which is internally mounted, or a touch-plate type that is externally mounted and suitable for hand operation.

* 3.3.3.5 Line cord. A flexible line cord and plug of one-piece construction shall be provided with three 18 (minimum) AWG conductors that terminate in a three-blade male plug that is described in ANSI C73-11. The exposed length of line cord and plug shall be not less than 5-1/2 feet.

3.4 Construction. The base shall house the electrical components, mechanical linkages, and holder for the cup receptacle. The top of the base shall have a cutout for mounting the cup receptacle to the holder. The bottom of the cup receptacle shall cover the cutout; or, as an alternate, the outer periphery of the cup receptacle shall have a rim overlapping the opening in the cutout. The front of the base shall have suitable cutouts to mount the knob-controlled rheostat, the dial plate and the on-off switch. If a touch-plate switch is provided (see 3.3.3.4), suitable cutouts shall be provided for top mounting the switch on the front edge of the base. The components shall be securely attached to the base. The line cord shall be connected to the electric controls through a cutout in the back of the base. Strain relief shall be provided at the cutout to secure the line cord and to prevent abrasion to the line cord

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insulation. The bottom of the base shall have a removable plate to permit access to the internal components. The plate shall be provided with ventilation holes. If an internally mounted microswitch with leaf actuator is supplied (see 3.3.3.4), suitable means shall be supplied to allow for manually adjusting the tension on the lead actuator. Two sponge strips shall be glued to the underside of the plate and shall run parallel to the front edge of the base. The bottom of the base shall be rectangular with an area of 25 to 30 square inches and no side less than 4-1/2 inches. The overall height of the laboratory mixer, including sponge strips and cup receptacle, shall not exceed 7 inches. All corners and edges of the base shall be rounded.

* 3.5 Performance. The laboratory mixer shall be suitable for mixing fluid material in test tubes from micro size up to and including 26 millimeters (mm.) diameter x 200 mm. long, and in vessels of various sizes and shapes. The mixing action of the laboratory mixer shall result from the motor rotation--through mechanical linkages--causing the cup receptacle to oscillate in an elliptical or circular orbit without revolving. The amplitude of oscillation shall be 1/16 to 1/8 inch. The mixing action in a test tube shall be accomplished by manually holding the tip of the tube, and inserting the tube in the cavity of the cup receptacle. With the top of the tube manually held in position, the bottom of the test tube shall move with the motion of the cup receptacle causing a whirlpool motion of the fluid material within the tube. The mixing action for vessels such as flasks, beakers, or odd-shaped containers shall be accomplished by manually holding such vessels against the side of the cup receptacle, the motion of the cup receptacle causing turbulence of the fluid material. The speed of the cup receptacle shall be regulated by the setting of the rheostat control. The changes in speed shall be stepless, and shall include 2100 to 2740 vibrations per minute with no load applied to the cup receptacle at 60 Hz AC. When operating on 50 Hz AC, a 5/6 reduction in speed from that at 60 Hz operation shall be acceptable.

The on-off switch shall control the operation of the laboratory mixer for either continuous or intermittent mixing. With the line plug connected to a power outlet, the laboratory mixer shall operate when the switch is in the "on" position. When the switch is in the "off" position, the motion of the cup receptacle shall be started or stopped by either of the pressure-actuating switches indicated in 3.3.3.4, and as indicated below:

a. The internally-mounted switch auromatically starting the cup receptacle in motion when the cup receptacle is loaded and stopping the motion when the load to the cup receptacle is removed.

b. The externally-mounted touch-plate when depressed automatically starting the cup receptacle into motion and stopping the motion when pressure is removed from the switch.

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As an alternate, a laboratory mixer with only intermittent mixing operation capability when switch is in the "on" position shall be acceptable.

The laboratory mixer shall not require more than 1/2 ampere line current for continuous operation. When tested as specified in 4.4.1, the laboratory mixer shall operate without any type of failure, and the sponge strips shall prevent the laboratory mixer from creeping or marring the surface on which the mixer is operating. There shall be no electrical breakdown when the laboratory mixer is subjected to the dielectric test specified in 4.4.2.

3.6 Finish. The exterior of the base shall have a baked-on enamel finish which shall be resistant to corrosion or scratching. The manufacturer's standard commercial colors shall be acceptable.

* 3.7 Identification marking. Each laboratory mixer shall be permanently and legibly marked with the NSN, contractor or purchase order number, name or registered trademark of the manufacturer, model number, serial number, and electrical characteristics.

3.8 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon written request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

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3.9 Workmanship. Workmanship shall be first class throughout. The laboratory mixer shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth by the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the cup receptacle, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

* 4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Table I and MIL-STD-105.

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Table I. For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-1	2.5

* 4.2.2 For tests. Tests shall be conducted in accordance with Table II and MIL-STD-105.

* Table II. For tests

Characteristic	Requirement	Test procedure	Inspection level
Performance	3.5	4.4.1	100%
Dielectric strength	3.5	4.4.2	100%
Durometer hardness	3.2.3	4.4.3	*

* A minimum of five specimens shall be tested for each lot of material in accordance with Fed. Std. No. 601. Acceptance number shall be zero.

4.3 Examination. The laboratory mixer shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with Table III.

Table III. Classification of defects

Categories	Defects
Major	
101	Unit not complete with all component parts.
102	Unit not free of cracks, tears, dents, and distortion.
103	Unit not free of wobbling and rocking when placed in a flat surface.
104	Sponge strips not adequately secured to bottom of base, and not free of cuts, tears or abrasions.
105	Unit not free of sharp edges and burrs.
106	Line cord not adequately secured within housing.
107	Line cord not free of cuts, tears and abrasions.
108	Blades of plug (line cord) not properly aligned.
109	Bottom plate not capable of being properly secured to base.
110	Components not usable and not capable of being properly secured to unit.

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Table III. Classification of defects (continued)

Categories	Defects *
Minor	
201	Identification marking not accurate, complete, legible and permanent.
202	Finish not free of scratches, chipping and blisters.
203	Unit not free of dirt, grease and foreign matter.

* Examinations shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The laboratory mixes shall be examined for defects in dimensions. Any dimension not within the specified limits shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

* 4.4.1 Performance. The rheostat shall be set to the low end of the speed range. With the line plug connected, the switch shall be set to the position for continuous operation, where applicable, and the rheostat speed control shall be slowly increased to maximum speed. The maximum speed shall be maintained for 30 seconds. The rheostat speed control shall be gradually decreased to minimum setting. With the switch in the position for intermittent operation, the pressure-actuating switch shall be activated and the rheostat speed control shall be slowly increased to maximum speed. The maximum speed shall be maintained for 30 seconds and then gradually decreased to minimum setting of the rheostat speed control.

4.4.2 Dielectric strength. With the line plug disconnected and the pressure-actuating switch deactivated, the laboratory mixer shall be subjected to a 60 Hz AC potential, gradually increasing to 900 volts, applied between the current-carrying parts and metal enclosure, and maintained at maximum value for a period of 1 minute.

4.4.3 Durometer hardness. Durometer hardness shall be determined by a shore durometer as specified in Fed. Std. No. 601, method 3021.

* 4.5 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirments of MIL-STD-129.

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* 5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C, as specified (see 6.1).

5.1.1 Unit of issue. Each shaking machine, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning. Shaking machines shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.2.2 Drying. Shaking machines shall be dried in accordance with MIL-P-116.

5.1.2.3 Preservative application. Preservatives shall not be used with the exception of preservatives applied to internal moving parts as deemed necessary by the contractor.

5.1.2.4 Unit packaging. Each unit shall be packaged in accordance with MIL-P-116, method IC-2.

5.1.3 Level C. Shaking machines shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A, B or C, as specified (see 6.1).

5.2.1 Level A. Four units, packaged as specified in 5.1.2, shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of PPP-B-636.

5.2.2 Level B. Four units, packaged as specified in 5.1.2, shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged shaking machines shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

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5.3 Marking. Unit packages and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type which is normally procured to meet the military requirements.

6.3 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-926-1290	SHAKING MACHINE, LABORATORY Single Cup, 115 volt, 50-60 Hz, AC

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6640-0969

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-36908A Shaking, Machine, Laboratory

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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